

Detecting Offensive Suppression and Evaluating Offensive Adaptation in NBA Games

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Abstract

Over the course of an NBA game, defenses adjust to take away play types that were efficient earlier, yet offenses are frequently observed continuing to run those actions rather than reallocating to alternatives the defense has left open. Aggregate season- and game-level statistics cannot detect this because it is a within-game, sequence-level phenomenon that disappears when possessions are pooled. Using a possession-level graph of 863,752 possessions across 5,234 games (2022–23 through 2025–26, regular season and playoffs), we define a suppression trigger as a play type whose second-half efficiency falls at least 0.25 points per possession (PPP) below its first half, with at least five possessions in each half. The central identification problem is that triggers are selected on elevated first-half efficiency, so a naive comparison confounds suppression with regression to the mean. We address this with a same-game, same-play-type matched-pair difference-in-differences estimator, balanced on first-half level via a caliper and inferred with cluster-robust standard errors and a cluster-level permutation test. Three findings follow. Suppression is real net of regression to the mean: suppressed play types lose roughly 0.41 PPP more into the second half than matched controls, an estimate stable across caliper choice, garbage-time exclusion, overtime handling, and game-level clustering. Teams do not effectively adapt: they continue the suppressed play at near-undiminished volume in the large majority of events, and substituting the initiating player does not recover efficiency. A counter-selection framework validated out-of-sample substantially outperforms continuation. We frame the framework as diagnostic—it detects suppression, quantifies its cost, and ranks counters—while emphasizing that pivot execution was never observed as a live intervention and no causal mechanism is identified.

1 Introduction

NBA offenses regularly keep running a play type after a defense has taken it away. Over the course of a game, a defense will adjust its coverage so that an action producing efficient offense in the first half no longer does in the second—and yet the offense, more often than not, continues calling that same action rather than shifting to something the defense has left available. This behavior is difficult to see with the tools most commonly applied to basketball analytics. Aggregate season- and game-level statistics collapse a game into summary rates, and in doing so they discard the very structure that makes this phenomenon visible: it is a within-game, sequence-level effect, in which what a defense did to a play type in one stretch shapes what happens when the offense returns to it. Pooled across a game or a season, the mid-game collapse of a single action disappears into an average.

This paper establishes three things. First, offensive suppression—the mid-game decline of a play type’s efficiency beyond ordinary variation—is real and measurable, and persists after accounting for regression to the mean and other same-game confounds. Second, offenses overwhelmingly fail to adapt to it: they continue running the suppressed play at nearly undiminished volume, and they do so even when a higher-scoring alternative already exists within the same game. Third, a counter-selection framework validated out-of-sample identifies responses that recover efficiency, and these validated counters substantially outperform continuation.

Measuring the first of these claims is harder than it appears, and the difficulty is worth stating at the outset. A play type is flagged as suppressed precisely because its efficiency fell—which means it was, by construction, performing unusually well beforehand. Anything performing unusually well tends to regress toward its typical level regardless of what the defense does. A naive comparison of a suppressed play’s decline therefore conflates genuine defensive suppression with ordinary regression to the mean, and cannot, on its own, tell the two apart. Separating them requires a design that holds the pre-decline level fixed across a suitable comparison; we develop that design in Section 3.

The practical stakes are concrete but bounded. If suppression is real, common, and costly, and if the information needed to respond to it is already present within the same game, then a coaching staff able to recognize suppression while a half remains—and to see which counters have historically survived the relevant matchup—would have decision support at the moment it matters. We frame this as a diagnostic contribution: the framework detects suppression, quantifies its cost, and ranks available counters,

but it does not observe teams executing those counters as live interventions, and it makes no causal claim about why a given defense adapted. What it offers is measurement of a recurring and largely unaddressed inefficiency, not proof of a remedy.

The remainder of the paper proceeds as follows. Section 2 relates the approach to prior work. Section 3 describes the data, the play-type classification and its calibration, the suppression-trigger definition, and the matched-pair identification strategy that separates suppression from regression to the mean, together with the robustness checks applied to it. Section 4 reports the three findings and the team-level heterogeneity around them. Section 5 synthesizes the findings, states what the framework can and cannot do, and outlines directions for future work. Section 6 concludes.

2 Related Work

Basketball analytics has largely been built on aggregate, possession-normalized measurement. The possession as the unit of efficiency, and points per possession as its currency, were formalized in the foundational performance-analysis literature [Oliver 2004; Kubatko et al. 2007], which established the offensive and defensive rating framework and the “four factors” decomposition that underlie nearly all modern team evaluation. Our efficiency metric inherits its possession-counting basis directly from this line of work. What that framework does not attempt, and does not need to attempt, is to resolve efficiency *within* a game at the level of a specific action over time: its unit is the game or the season, and a play type’s mid-game collapse is averaged away by construction.

A second line of work estimates player value across games by adjusting for context. Regularized adjusted plus-minus and its relatives [Sill 2010] regress margin or efficiency onto on-court indicators, using regularization and out-of-sample validation to produce stable cross-game player ratings. These methods answer a different question from ours—the aggregate, season-scale contribution of a player—and operate at the level of lineups and games rather than sequential possessions of a given action. They are complementary to, not competing with, a within-game suppression detector.

Closest to the present work in spirit is the tracking-data literature on continuous possession valuation, most prominently expected possession value [Cervone et al.], which models the evolving value of a possession from spatiotemporal player-tracking data and can value individual decisions in real time. That work and ours share the premise that possession-level, sequence-

aware modeling recovers signal that aggregates discard. They differ in two respects that define our contribution. First, expected possession value depends on proprietary optical-tracking data; our detector operates on public play-by-play, which broadens applicability at the cost of spatial detail. Second, expected possession value is a continuous valuation of decisions as they occur, whereas we target a specific, discrete phenomenon—the mid-game suppression of a play type and the offense’s response to it—and pair its detection with an out-of-sample-validated ranking of counters.

Our contribution is positioned accordingly. We operate at the within-game, possession-sequence level, like the tracking-based valuation literature, but from public play-by-play, like the aggregate-rating tradition. We do not claim priority over or improvement upon these approaches; they address distinct questions and rest on different data. What is, to our knowledge, not jointly addressed by the existing lines above is the task of detecting mid-game play-type suppression, distinguishing it from regression to the mean, and evaluating whether teams adapt—which is the gap this paper fills.

3 Methods

3.1 Data

All analyses draw on a possession-level graph database (Neo4j) covering the 2022–23 through 2025–26 NBA seasons, regular season and playoffs combined. The graph contains 863,752 classified possessions across 5,234 games and 30 teams. Each possession record carries the offensive team, an inferred play type, points scored, period, and a pre-possession score margin, the last of which is populated on all classified possessions. Points per possession (PPP) is the primary efficiency metric throughout; all reported averages are arithmetic row-means (each qualifying unit contributes one equally weighted row), never possession-weighted pooled means.

3.2 Possession classification and PNR calibration

Play types are inferred from play-by-play (PBP) event streams rather than read from a labeled feed. The inferred taxonomy includes PNR, SPOT_UP, DRIVE, PULL_UP, POST_UP, CUT, TRANSITION, PUTBACK, and an OTHER fallback.

Pick-and-roll (PNR) requires a dedicated calibration pass. The NBA V3 PBP feed systematically under-labels it: assist flags appear only on made

shots, and screen or pick language never appears in the V3 description text, so a PNR ball-handler who misses, or who scores unassisted off a screen, is never captured by the inferrer’s PNR rules. Left uncorrected, inferred PNR runs near 4 percent of possessions against a true rate of roughly 20–25 percent. The calibration corrects this at ingest time, before possessions are written to the graph, by promoting eligible PULL_UP and OTHER possessions to PNR up to a per-team target derived from a team’s reference PNR shot-share scaled by a factor of 2.5, capped at 33 percent of half-court possessions. Promotion is scored (player historical PNR frequency times a confidence-based prior), applied independently within each half to preserve intra-game asymmetry, and bounded so that no more than half of a given bucket’s PULL_UP possessions are drained in one pass. Because calibration mutates the stored play type, it is upstream of all downstream trigger computation; it is not a pregame-only or report-time adjustment. Calibration is validated only at the population level, against league- and player-level target shares; there is no possession-by-possession ground-truth validation, and this limitation is stated again in Section 5.

3.3 Notation and the suppression trigger

Let a possession belong to team t (offense), game g , and play type p . Periods 1–2 constitute the first half (H1) and periods 3–4 the second half (H2); overtime periods are excluded from both. For each (t, p, g) let $n_1(t, p, g)$ and $n_2(t, p, g)$ be the number of half-court possessions of type p run by t in each half, and let

$$\text{PPP}_h(t, p, g) = \frac{1}{n_h(t, p, g)} \sum_{i \in \mathcal{P}_h(t, p, g)} \text{ppp}_i, \quad h \in \{1, 2\}, \quad (1)$$

be the mean points per possession over that half’s possession set $\mathcal{P}_h$.

Definition 1 (Suppression trigger). *A play type p run by team t in game g is a suppression trigger if*

$$n_1(t, p, g) \geq 5, \quad n_2(t, p, g) \geq 5, \quad \text{PPP}_1(t, p, g) - \text{PPP}_2(t, p, g) \geq 0.25. \quad (2)$$

The first two conditions ensure both halves are estimable; the third requires a first-to-second-half efficiency decline of at least 0.25 PPP. Applied to the graph, this yields the counts in Table 1.

Table 1: Dataset and trigger summary.

Quantity	Value
Possessions	863,752
Games	5,234
Seasons	2022–23 to 2025–26 (RS + PO)
Suppression triggers	11,064 (RS 10,418; PO 646)
Trigger rule	$n_1, n_2 \geq 5$; $\text{PPP}_1 - \text{PPP}_2 \geq 0.25$

3.4 Matched-pair identification

The central identification problem is regression to the mean. A unit qualifies as a trigger precisely because its second-half efficiency fell, which means its first-half efficiency was elevated by construction; higher first-half levels regress more strongly regardless of any defensive adjustment. A naive comparison therefore attributes ordinary mean reversion to suppression.

For a triggered unit $T = (t, p, g)$, the control $C = (t', p, g)$ is the opposing team t' in the same game running the same play type, subject to the symmetric sample floors $n_1(C) \geq 5$ and $n_2(C) \geq 5$. Define each unit’s drop and the pairwise difference-in-differences (the *excess*):

$$\begin{aligned}
D(U) &= \text{PPP}_1(U) - \text{PPP}_2(U), \\
\Delta_i &= D(T_i) - D(C_i) \\
&= [\text{PPP}_1(T_i) - \text{PPP}_2(T_i)] \\
&\quad - [\text{PPP}_1(C_i) - \text{PPP}_2(C_i)].
\end{aligned} \tag{3}$$

The estimator is the pair mean

$$\hat{\Delta} = \frac{1}{N} \sum_{i=1}^N \Delta_i. \tag{4}$$

A pair is retained only if the two units are matched on first-half level:

$$|\text{PPP}_1(T_i) - \text{PPP}_1(C_i)| \leq c, \quad c = 0.10 \text{ (primary)}. \tag{5}$$

The naive and corrected estimators are identical in form and differ only in the retained pair set: $\hat{\Delta}_{\text{naive}}$ imposes only $n_1(C) \geq 5$ with no caliper ($c = \infty$), whereas $\hat{\Delta}_{\text{corrected}}$ imposes $c = 0.10$ and $n_2(C) \geq 5$.

3.5 Inference

Let clusters be indexed by game, $g = 1, \dots, G$, with residuals $r_i = \Delta_i - \hat{\Delta}$ and per-cluster residual sums $S_g = \sum_{i \in g} r_i$. The CR1 cluster-robust variance of the mean is

$$\widehat{\text{Var}}(\hat{\Delta}) = \frac{G}{G-1} \cdot \frac{1}{N^2} \sum_{g=1}^G S_g^2, \quad \text{SE} = \sqrt{\widehat{\text{Var}}(\hat{\Delta})}, \quad (6)$$

with the 95% interval $\hat{\Delta} \pm t_{0.975, G-1} \text{SE}$. The design effect and effective sample size are

$$\text{DEFF} = \left(\frac{\text{SE}_{\text{cluster}}}{\text{SE}_{\text{iid}}} \right)^2, \quad n_{\text{eff}} = \frac{N}{\text{DEFF}}. \quad (7)$$

Under $H_0 : E[\Delta_i] = 0$, the signs of Δ_i are exchangeable at the cluster level. For each of $B = 5,000$ permutations we draw an independent sign $s_g \in \{-1, +1\}$ with equal probability per cluster, apply it to all pairs in that cluster, and recompute $\hat{\Delta}^{(b)} = \frac{1}{N} \sum_i s_{g(i)} \Delta_i$. The one-sided p -value is

$$p = \frac{1}{B} \sum_{b=1}^B \mathbf{1}[\hat{\Delta}^{(b)} \geq \hat{\Delta}]. \quad (8)$$

3.6 Robustness checks

The locked specification was re-estimated under a caliper sweep (c from 0.05 to 0.30), garbage-time exclusion of second-half possessions with absolute score margin above 15, 20, and 25 points, game-level clustering, and correction of the half boundary to exclude overtime from both halves. Results appear in Table 3.

4 Results

4.1 Offensive suppression exceeds regression to the mean

On the first-half-balanced matched-pair design (caliper $c = 0.10$, symmetric sample floors), the excess is +0.408 PPP (cluster-robust 95% CI [0.378, 0.438]; $N = 1,136$ pairs across $G = 848$ game clusters; permutation $p < 0.0002$). Triggered teams declined from 1.305 PPP in the first half to 0.751 in the second, while their matched controls declined from 1.298 to 1.153 (Table 4). A play type identified as suppressed thus loses about 0.41 PPP more into the

Table 2: Matched-pair difference-in-differences—naive vs. corrected. Excess is $\hat{\Delta}$ in PPP; CI is cluster-robust on game; p is the cluster-level permutation p -value. N = matched pairs; G = game clusters; $n_2(C) \geq 5$ = control second-half sample floor.

Estimator	c	$n_2(C)$	N	G	H1 gap	Excess	95% CI	p
Naive	∞	no	9,111	4,342	+0.308	+0.625	[0.611, 0.639]	< 0.0002
Corrected	0.10	yes	1,136	848	+0.007	+0.408	[0.378, 0.438]	< 0.0002

Table 3: Robustness of the corrected estimate. Caliper sweep and garbage-time exclusion; all rows retain $n_2(C) \geq 5$ and overtime exclusion.

Condition	Excess $\hat{\Delta}$	95% CI
Caliper 0.05	+0.403	[0.362, 0.445]
Caliper 0.10 (primary)	+0.408	[0.378, 0.438]
Caliper 0.15	+0.397	[0.373, 0.421]
Caliper 0.20	+0.398	[0.377, 0.418]
Caliper 0.30	+0.412	[0.395, 0.429]
Blowout $ \text{margin} \leq 15$	+0.403	[0.370, 0.437]
Blowout $ \text{margin} \leq 20$	+0.412	[0.380, 0.443]
Blowout $ \text{margin} \leq 25$	+0.416	[0.385, 0.448]

second half than a comparable control, after netting out regression to the mean. The naive estimator, by contrast, returns +0.625 PPP on 9,111 pairs with a first-half imbalance of +0.308 PPP (Table 2); balancing on first-half level collapses that imbalance to +0.007 PPP and the estimate to +0.408.

The estimate is stable across every robustness condition (Table 3): it ranges only from +0.397 to +0.412 across the caliper sweep, equals +0.403 / +0.412 / +0.416 under garbage-time exclusion, is left effectively unchanged by game-level clustering (design effect ≈ 1.0), and shifts by less than the confidence-interval half-width when overtime is excluded from the half definition. This result is associative: the matched pair removes regression to the mean and same-game confounds but does not identify the defensive mechanism.

4.2 Teams do not effectively adapt

Offenses continue running the suppressed play type at nearly undiminished volume: 97 percent (playoffs) and 98 percent (regular season) of first-half

Table 4: Component half efficiencies at the primary specification ($c = 0.10$).

Unit	PPP ₁	PPP ₂	Drop D
Triggered	1.305	0.751	−0.554
Control	1.298	1.153	−0.145
Difference ($\hat{\Delta}$)			+0.408

Table 5: Non-recovery via initiator substitution (2025–26 RS).

Group	n	Drop D
Same primary initiator	825	0.600
Swapped primary initiator	1,732	0.639
Swap – same		+0.039 ($p = 0.002$, $d = 0.12$)

volume on the suppressed action carries into the second half, and teams keep running the play in 93 percent of suppression events. Continuing costs a per-trigger mean of 0.63 PPP relative to the first half, or roughly 5.4 points per trigger in the playoffs and 5.5 in the regular season; this cost is insensitive to the first-half efficiency floor used in the continuation analysis (regular-season cost moves 5.70 to 5.55 when the floor is removed).

Substituting the initiating player does not recover efficiency (Table 5). Among 2,557 classified regular-season triggers, 68 percent ($n = 1,732$) swapped the primary initiator and 32 percent ($n = 825$) retained it; the retained group dropped 0.600 and the swapped group dropped 0.639. The swap-minus-retain difference is +0.039 PPP ($p = 0.002$, Cohen’s $d = 0.12$)—a negligible effect: distinguishable from zero on this sample but far too small to constitute recovery, and directionally opposite to adaptation. The second-half collapse is therefore a property of the play type under the deployed defense, not of the individual initiator. These analyses use held-out regular-season and playoff cohorts and row-mean PPP per trigger.

4.3 No systematic pivots occur even when a superior option existed

Across 1,539 analyzed triggers, 1,466 (95.3 percent) had a higher-scoring play type already available within the same game, yet no team systematically shifted volume away from the suppressed play toward the superior one (zero significant pivots). The failure to adapt is not attributable to the absence of

Table 6: Response ladder (row-mean PPP).

Response	Playoffs	Regular season
Continue suppressed play	0.853	0.839
Worst counter (AVOID)	0.923	1.001
Best counter (BEST)	1.298	1.423
BEST – continue (primary)	+0.445	+0.584
BEST – AVOID (tier gap)	+0.375	+0.422

a viable alternative: the alternative existed and went unused. This analysis uses the held-out regular-season cohort with row-mean PPP per trigger.

4.4 Counters outperform continuation

On the response ladder (Table 6), continuing the suppressed play scores 0.853 PPP (playoffs) and 0.839 (regular season). Every counter tier exceeds continuation, and the matchup-ranked best counter scores 1.298 and 1.423. The primary quantity—best counter versus continuation—is +0.445 PPP (playoffs) and +0.584 (regular season). As a distinct tier-validation statistic, the gap between the best and worst counter tiers is +0.375 (playoffs) and +0.422 (regular season); this measures whether the ranking separates good counters from bad and is not the same as the best-versus-continuation quantity.

The essential caveat is that continuation PPP and counter PPP are measured on different possessions within the same game, not on a paired counterfactual. This is an opportunity-cost comparison; it does not establish that a team executing the counter would have realized the observed counter efficiency.

4.5 Team-level heterogeneity

Suppression cost and counter availability vary substantially across teams. Documented cases include New York (68.8 percent of triggers with no sufficient counter available; 41 percent win rate) and Denver (71.8 percent; 60 percent win rate), against a counterexample in Indiana (18.6 percent best-counter follow rate; 71.4 percent win rate across seven series-directions). Counter-selection quality and winning are therefore not tightly coupled at the team level: teams with poor counter availability can still win, and disciplined counter-following does not guarantee success.

5 Discussion

5.1 Synthesis

Three findings cohere into a single account of in-game offensive adaptation. First, offensive suppression is a real and measurable defensive phenomenon rather than a statistical artifact: after matching triggered play types to same-game controls at comparable first-half efficiency and correcting for regression to the mean, garbage time, overtime, and within-game clustering, suppressed actions still lose roughly 0.41 PPP more into the second half than their controls. Second, offenses overwhelmingly fail to respond—they continue the suppressed play at near-undiminished volume in 93 percent of events, and even when a higher-scoring alternative already exists in the same game (95.3 percent of analyzed triggers), no team systematically reallocates toward it. Third, a counter-selection framework validated out-of-sample separates responses that recover efficiency from those that do not, with the matchup-ranked best counter outperforming continuation by +0.445 PPP (playoffs) and +0.584 (regular season).

Together these results imply that a large, recurring, and largely un-addressed efficiency loss exists in NBA offenses, and that the information needed to avoid it is present within the same game. The natural coaching implication is that recognizing suppression earlier and reallocating away from the suppressed action would be valuable. This must be stated carefully: the counter comparisons are opportunity-cost differences measured on distinct possessions, not paired counterfactuals, and no team was observed executing a recommended pivot as a live intervention. The framework establishes that continuation is costly and that better-scoring alternatives coexist with the suppressed play; it does not establish that a team switching to those alternatives would realize the observed counter efficiency against a defense already shown capable of adapting.

5.2 What the framework can and cannot do

The framework can detect suppression events with their timing, quantify the cost of ignoring suppression at roughly 5.4 points per trigger, compare teams on their potential response given available counters, and flag talent-compensation cases where counter availability and outcomes diverge.

Its scope limits are genuine, not rhetorical. It cannot determine whether pivoting is executable, because pivot recommendations are counterfactual rather than observed interventions. It cannot explain why a defense adapted: the underlying data contain no coverage labels, no defensive assignments,

and no scheme tags, so the mechanism behind any given suppression is unidentified. It cannot make causal claims—every finding is associative by design. And it cannot identify executable counter identity below the play-type level; the recommendations operate on play types, not on the specific personnel actions a staff would ultimately call.

5.3 Practical implications

For a coaching staff, the practical value would lie in converting a signal currently recognized late or not at all into one available with a full half remaining to act. Real-time suppression detection would flag that a play type’s second-half efficiency has collapsed beyond normal variation while there is still time to reallocate, and a ranked-counter view would indicate which alternatives have historically survived against the relevant matchup. Given that continuation is both common and costly and that superior alternatives typically coexist with the suppressed action, the immediate operational benefit is decision support at the moment of choice rather than post-hoc explanation. The magnitude of any realized gain remains bounded by the execution caveat above.

5.4 Future work

Several questions follow directly from the present scope limits. First, whether the signal changes behavior when teams have access to it—the current analysis observes teams that did not use suppression detection. Second, whether suppression can be predicted before it occurs, rather than detected only after the second-half collapse has begun. Third, whether suppression targeting varies within a game and can be detected earlier than the half-level trigger used here, shortening the window between onset and actionable recognition. A fourth direction, implied by the mechanism gap, is incorporating coverage or assignment information, where obtainable, to move from detecting that suppression occurred toward explaining why.

6 Conclusion

We introduce a within-game, possession-sequence framework that detects the mid-game suppression of offensive play types from public play-by-play data, distinguishes it from regression to the mean via same-game matched-pair difference-in-differences, and evaluates whether offenses adapt. Suppression is real net of regression to the mean, with suppressed play types

losing about 0.41 PPP more into the second half than matched controls, an estimate robust to caliper choice, garbage-time exclusion, overtime handling, and game-level clustering. Offenses largely fail to adapt, continuing the suppressed play even when a superior alternative is already available, and validated counters substantially outperform continuation. The contribution is diagnostic: the framework measures a recurring and largely unaddressed inefficiency and ranks the counters available to address it, while stopping short of claiming a validated intervention or a causal mechanism. Whether teams that act on the signal can convert it into realized efficiency is the central question we leave to future work.

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